

Ex. 14. Of two numbers, 4 times the smaller one is less than 3 times the larger one by 5. If the sum of the numbers is larger than 6 times their difference by 6, find the two numbers.

Sol. Let the numbers be x and y , such that $x > y$.

$$\text{Then, } 3x - 4y = 5 \quad \dots(i)$$

$$\text{and } (x + y) - 6(x - y) = 6 \Rightarrow -5x + 7y = 6 \quad \dots(ii)$$

Solving (i) and (ii), we get : $x = 59$ and $y = 43$.

Hence, the required numbers are 59 and 43.

Ex. 15. The ratio between a two-digit number and the sum of the digits of that number is 4 : 1. If the digit in the unit's place is 3 more than the digit in the ten's place, what is the number? (S.B.I.P.O., 2005)

Sol. Let the ten's digit be x . Then, unit's digit = $(x + 3)$.

Sum of the digits = $x + (x + 3) = 2x + 3$. Number = $10x + (x + 3) = 11x + 3$.

$$\therefore \frac{11x+3}{2x+3} = \frac{4}{1} \Leftrightarrow 11x + 3 = 4(2x + 3) \Leftrightarrow 3x = 9 \Leftrightarrow x = 3.$$

Hence, required number = $11x + 3 = 36$.

Ex. 16. A number consists of two digits. The sum of the digits is 9. If 63 is subtracted from the number, its digits are interchanged. Find the number.

Sol. Let the ten's digit be x . Then, unit's digit = $(9 - x)$.

$$\text{Number} = 10x + (9 - x) = 9x + 9.$$

Number obtained by reversing the digits = $10(9 - x) + x = 90 - 9x$.

$$\therefore (9x + 9) - 63 = 90 - 9x \Leftrightarrow 18x = 144 \Leftrightarrow x = 8.$$

So, ten's digit = 8 and unit's digit = 1.

Hence, the required number is 81.

Ex. 17. In a three-digit number, the digit in the units place is four times the digit in the hundreds place. If the digits in the units place and the tens place are interchanged, the new number so formed is 18 more than the original number. If the digit in the hundreds place is one-third of the digit in the tens place, what is the original number? (L.I.C., 2005)

Sol. Let the digit in the hundreds place be x .

Then, digit in the units place = $4x$. And, digit in the tens place = $3x$.

$$\text{Original number} = 100x + 10 \times 3x + 4x = 100x + 30x + 4x = 134x.$$

Number formed on interchanging the unit's and ten's digits

$$= 100x + 10 \times 4x + 3x = 143x.$$

$$\therefore 143x - 134x = 18 \Leftrightarrow 9x = 18 \Leftrightarrow x = 2.$$

Hence, original number = $134x = (134 \times 2) = 268$.

Ex. 18. If the digits of a two-digit number are interchanged, the number formed is greater than the original number by 45. If the difference between the digits is 5, what is the original number? (Bank P.O., 2009)

Sol. Since the number formed by interchanging the digits is greater so the ten's digit of the original number is smaller than the unit's digit.

Let the ten's digit be x . Then, unit's digit = $x + 5$.

$$\text{Original number} = 10x + (x + 5) = 11x + 5.$$

Number formed on interchanging the digits = $10(x + 5) + x = 11x + 50$.

$$\therefore (11x + 50) - (11x + 5) = 45 \Rightarrow 45 = 45, \text{ which is independent of } x.$$

Hence, the number cannot be determined from the given data.

Ex. 19. A fraction becomes $\frac{2}{3}$ when 1 is added to both its numerator and denominator. And, it becomes $\frac{1}{2}$ when 1 is subtracted from both the numerator and denominator. Find the fraction.

Sol. Let the required fraction be $\frac{x}{y}$. Then,

$$\frac{x+1}{y+1} = \frac{2}{3} \Rightarrow 3x - 2y = -1 \quad \dots(i) \text{ and } \frac{x-1}{y-1} = \frac{1}{2} \Rightarrow 2x - y = 1 \quad \dots(ii)$$

Solving (i) and (ii), we get : $x = 3, y = 5$.

$$\therefore \text{Required fraction} = \frac{3}{5}.$$